OBJECTIVE WORKSHEET

Acids, Bases and Salts

Science — Chemistry · Class 7 · Max Marks: 25 · Time: 45 min

Section A — Multiple Choice Questions (1 × 8 = 8)

1. A solution turns blue litmus red but has no effect on red litmus. The solution is:

A. neutral
B. basic
C. distilled water
D. acidic

2. Turmeric paste on a white cloth turns red where soap is rubbed. This shows that soap solution is:

A. neutral
B. basic
C. salty
D. acidic

3. Equal volumes of an acid and a base are mixed and the mixture then changes neither red nor blue litmus. The mixture contains:

A. salt and water
B. only acid
C. a stronger acid
D. only base

4. Which pair correctly matches an acid with its natural source?

A. tartaric acid – ant sting
B. citric acid – vinegar
C. lactic acid – curd
D. acetic acid – lemon

5. A gardener adds slaked lime (a base) to a field. The soil was most likely:

A. perfectly neutral
B. too basic
C. too salty
D. too acidic

6. Rubbing baking soda on an ant's sting relieves the pain because the baking soda:

A. neutralises the acid injected by the ant
B. cools the skin by evaporation
C. colours the skin
D. adds more acid

7. Solution P turns red litmus blue, Q turns blue litmus red and R changes neither. P, Q and R are respectively:

A. neutral, basic, acidic
B. acidic, basic, neutral
C. basic, acidic, neutral
D. all neutral

8. During the neutralisation of an acid by a base, the temperature of the mixture usually:

A. falls sharply
B. rises (heat is given out)
C. drops to zero
D. stays exactly the same

Section B — Fill in the Blanks (1 × 5 = 5)

1. An acid turns ______ litmus red.

2. Bases turn red litmus ______.

3. The reaction between an acid and a base is called ______.

4. Milk of magnesia, taken to relieve stomach acidity, is a mild ______.

5. China rose indicator turns ______ with an acidic solution.

Section C — True or False (1 × 5 = 5)

1. All acids are dangerous and are never found in the food we eat. (True / False)

2. Distilled water has no effect on either red or blue litmus. (True / False)

3. A solution that turns china rose indicator green is acidic. (True / False)

4. The salt formed on neutralising any acid with any base is always common salt (sodium chloride). (True / False)

5. Bases generally feel soapy to the touch and taste bitter. (True / False)

Section D — Match the Columns (1 × 4 = 4)

Column A	Column B
1. Acetic acid	a) curd
2. Formic acid	b) vinegar
3. Citric acid	c) lemon and orange
4. Lactic acid	d) ant's sting

Section E — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. **Assertion (A):** Toothpastes are usually basic in nature.
   Reason (R): They neutralise the acids formed in the mouth that cause tooth decay.
2. **Assertion (A):** Acidic factory waste is treated with basic substances before being released into rivers.
   Reason (R): The untreated acidic waste would harm the plants and animals living in the water.
3. **Assertion (A):** Lemon juice turns blue litmus red.
   Reason (R): Lemon juice is basic in nature.

Answer Key

Multiple Choice Questions

1. D **2.** B **3.** A **4.** C **5.** D
6. A **7.** C **8.** B

Fill in the Blanks

1. blue
2. blue
3. neutralisation
4. base (antacid)
5. dark pink (magenta)

True or False

1. False
2. True
3. False
4. False
5. True

Match the Columns

1-b, 2-d, 3-c, 4-a

Assertion–Reason

1. A
2. A

3. C